Name of the Experiment: CALCULATION OF MEASURES OF CENTRAL TENDENCY

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The following frequency distribution refers to the monthly income (100 tk) of 120 farmers.

	5-10	10-15	15-20	20-25	25-30	30-35	35-40
	5	10	30	45	20	8	2

(i) Calculate: (a) Anithmetic mean (b) Geometric mean (c) Harmonic mean (d) median (e) Mode (f) Q_1, Q_2, D_5 and P_{90}

(ii) Show that $Me = Q_2 = D_E = P_{50}$

(iii) Calculate the median and Q_1, P_{90} from the appropriate graph

(iv) Draw the histogram and locate mode.

Solution: The following table is constructed for calculation:

Class interval	Frequencies	Cumulative frequency	Mid value (x_i)	$f_i x_i$	$\log x_i$	$f_i \log x_i$	f_i / x_i
5-10	5	5	7.5	37.5	1.575	4.375	0.666 - 3
10-15	10	15	12.5	125	1.196	10.96	0.8 - 2
15-20	30	45	17.5	525	1.243	37.29	1.714 - 1
20-25	45	90	22.5	1012.5	1.352	60.84	2 0
25-30	20	110	27.5	550	1.439	28.78	0.727 1
30-35	8	118	32.5	260	1.511	12.088	0.246 2
35-40	2	120	37.5	75	1.574	3.148	0.533 3
Total	N=120			2585		157.481	6.206

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a) Arithmetic mean:

We know that Arithmetic mean $= \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$

$= \frac{2585}{120} = 21.54$

Comment: The monthly income (1000tk) of farmers in arithmetic mean is 21.54

(b) Geometric mean:

We know that Geometric mean $= \text{Antilog}\left\{\frac{1}{N}\sum_{i=1}^{n} f_i \log x_i\right\}$

$= \text{Antilog}\left\{\frac{157.481}{120}\right\}$

$= 20.528$

Comment: The monthly income (1000tk) of farmers in geometric mean is 20.528

(c) Harmonic mean:

We know that harmonic mean $= \frac{1}{\frac{1}{N}\sum_{i=1}^{n} f_i/x_i}$

$= \frac{1}{1/120 \times 6.206}$

$= 19.336$

Comment: The monthly income (1000tk) of farmers in harmonic mean is 19.336

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(d) Median: We know that,

$$M_e = L + \frac{\frac{N}{2} - f'_m}{f_m} \times h$$

$\frac{N}{2} = \frac{120}{2}$ = 60th observation

Lies in the class interval (20-25)

(20-25) is the median class

$$M_e = 20 + \frac{60-45}{45} \times 5$$

$$= 21.67$$

Where,

L = The lower limit of median class = 20

N = Total No of observation

F'm = Cumulative frequency of pre-median class = 45

Fm = Frequency of median class 45

h = Class interval

Comment: The monthly income (1000tk) of farmers in median is 21.67

(e) Mode: We know that,

$$M_o = L + \frac{f_o - f_1}{2 f_o . f_1 - f_2} \times h$$

Here, (20-25) is the modal class. Because maximum frequency.

$$M_o = 20 + \frac{45-30}{2 \times 45 - 30 - 20} \times 5$$

$$= 20 + 1.875$$

$$= 21.875$$

L = lower limit of the modal class = 20

f_1 = Frequency of modal class = 45

f_0 = Frequency of pre modal class = 30

f_2 = Frequency of post modal class = 20

h = class interval = 5

Comment: The monthly income (1000 tk) of farmers in mode (m_o) is 21.875

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(f) $Q_1 : \frac{N}{4} = \frac{120}{4} = 30$th observation lies in the class (15-20)

(15-20) is the lower quartiles (Q_1) class

$$Q_1 = L_1 + \frac{\frac{i \times N}{4} - F_i}{f_i} \times h$$

$$= 15 + \frac{1 \times 30 - 15}{30} \times 5$$

$$= 17.5$$

$L_1 = 15$

$i = 1$

$\frac{N}{4} = 30$

$F_i' = 15$

$f_i = 30$

$h = 5$

$Q_1 = ?$

Again, $\frac{2N}{4} = \frac{2 \times 120}{4} = 60$th observation lies in the class interval (20-25)

(20-25) is the mid quartile Q_2 class

$$Q_2 = L_2 + \frac{\frac{2 \times N}{4} - f_i'}{f_i} \times h$$

$$= 20 + \frac{60 - 45}{45} \times 5$$

$$= 21.67$$

$L_2 = 20$

$\frac{2N}{4} = 60$

$f_i' = 45$

$f_i = 45$

$h = 45$

$Q_2 = ?$

Comment: The monthly income (100 ötk) of farmers in Q_1 and Q_2 is 17.5 and 21.67 respectively.

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(g) Computation of Deciles: $D_5 : \frac{5N}{10} = \frac{5 \times 120}{10} = 60th$

Observation lies in (20-25)

∴ (20-25) is the 5th declies (D_5) class

$$D_5 = L + \frac{\frac{5N}{10} - f_j'}{f_j} \times h$$

$$= 20 + \frac{60 - 45}{45} \times 5$$

$$= 21.67$$

Where,
$L_5 = 20$
$\frac{5N}{10} = 60$
$F_j' = 45$
$f_j = 45$
$h = 5$
$D_5 = ?$

Comment: the monthly income (1000 tk) of farmers is 5th deciles (D_5) is 21.67

(h) Computation of percentiles: $P_{90} : \frac{90N}{100} = \frac{90 \times 120}{100} = 108th$

Observation lies in (25-30)

(25-30) is the 90th percentiles P_{90} class

$$P_{90} = L_{90} + \frac{\frac{90N}{120} - f'_K}{f_K} \times h$$

$$= 25 + \frac{108 - 90}{20} \times 5$$

$$= 29.5$$

Where,
$L_{90} = 25$
$\frac{90N}{100} = 108$
$f'_K = 90$
$f_K = 20$
$h = 5$
$P_{90} = ?$

Comment: The monthly income (1000 tk) of farmers in 90th percentiles is 29.5

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(i) $P_{50} = \frac{50 \times N}{100} = \frac{50 \times 120}{100} = 60$th observation lie in (20-25)

(20-25) is the 50th percentiles P_{50} class.

$$P_{50} = L_{50} + \frac{\frac{50N}{100} - f'_k}{f_k} \times h$$

$$= 20 + \frac{60 - 45}{45} \times 5$$

$$= 21.67$$

Where,

$L_{50} = 20$

$\frac{5N}{100} = 60$

$F'_k = 45$

$F_k = 45$

$h = 5$

$P_{50} = ?$

From the above calculation we get

$M_e = 21.67$ $Q_2 = 21.67$

$D_5 = 21.67$ $P_{50} = 21.67$

So, we can say that

$$M_e = Q_2 = D_5 = P_{50}$$

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Figure No.: Page No.: Date:

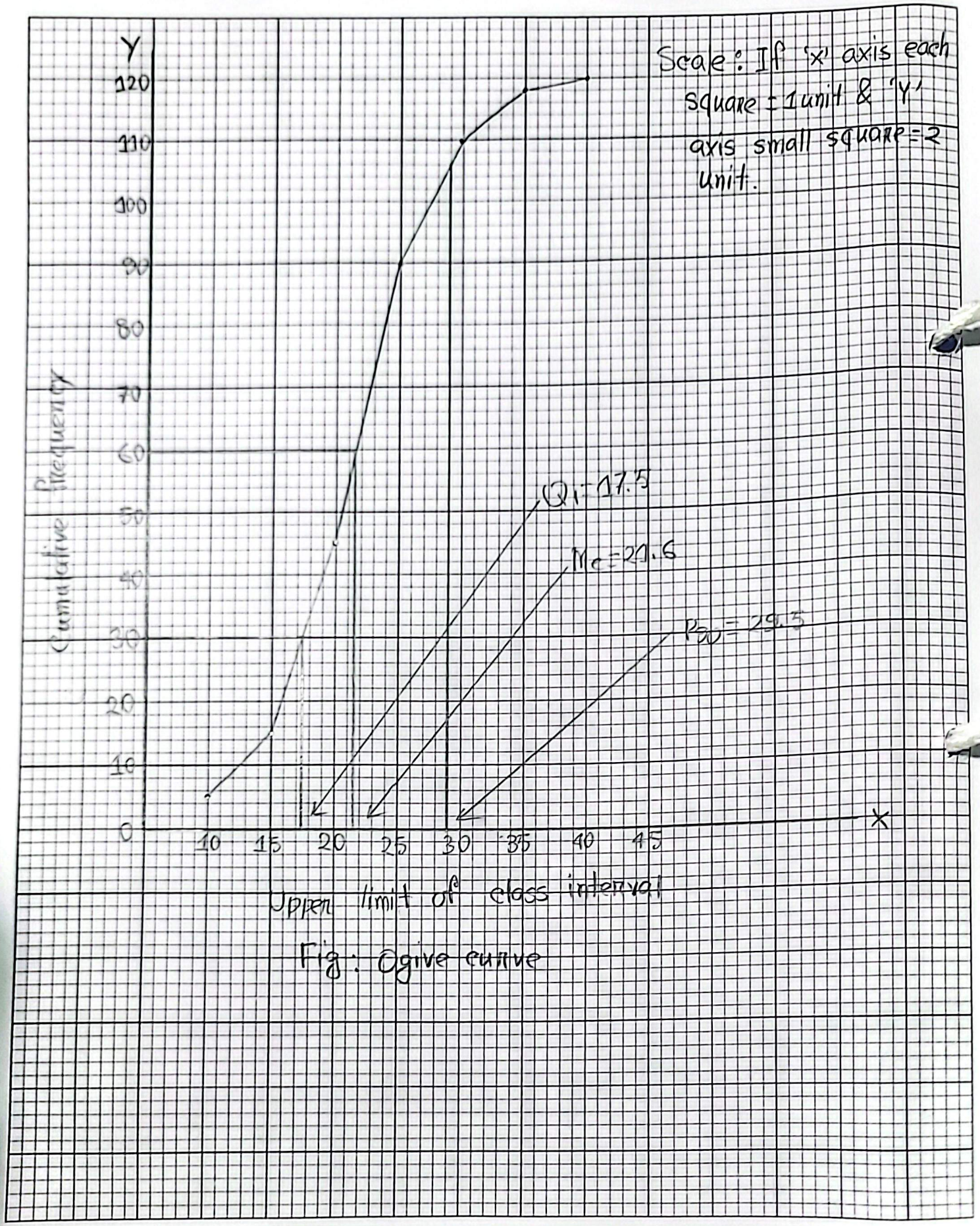


Fig: Ogive curve

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Comment: In the graph paper 'x' axis show the height values of class interval and each small square = 1 unit

The 'y' axis show the cumulative frequency and here each small square = 2 units.

In the ogive curve show that Q, Me, P_{90} is 17.5 21.67 and 29.5 respectively.

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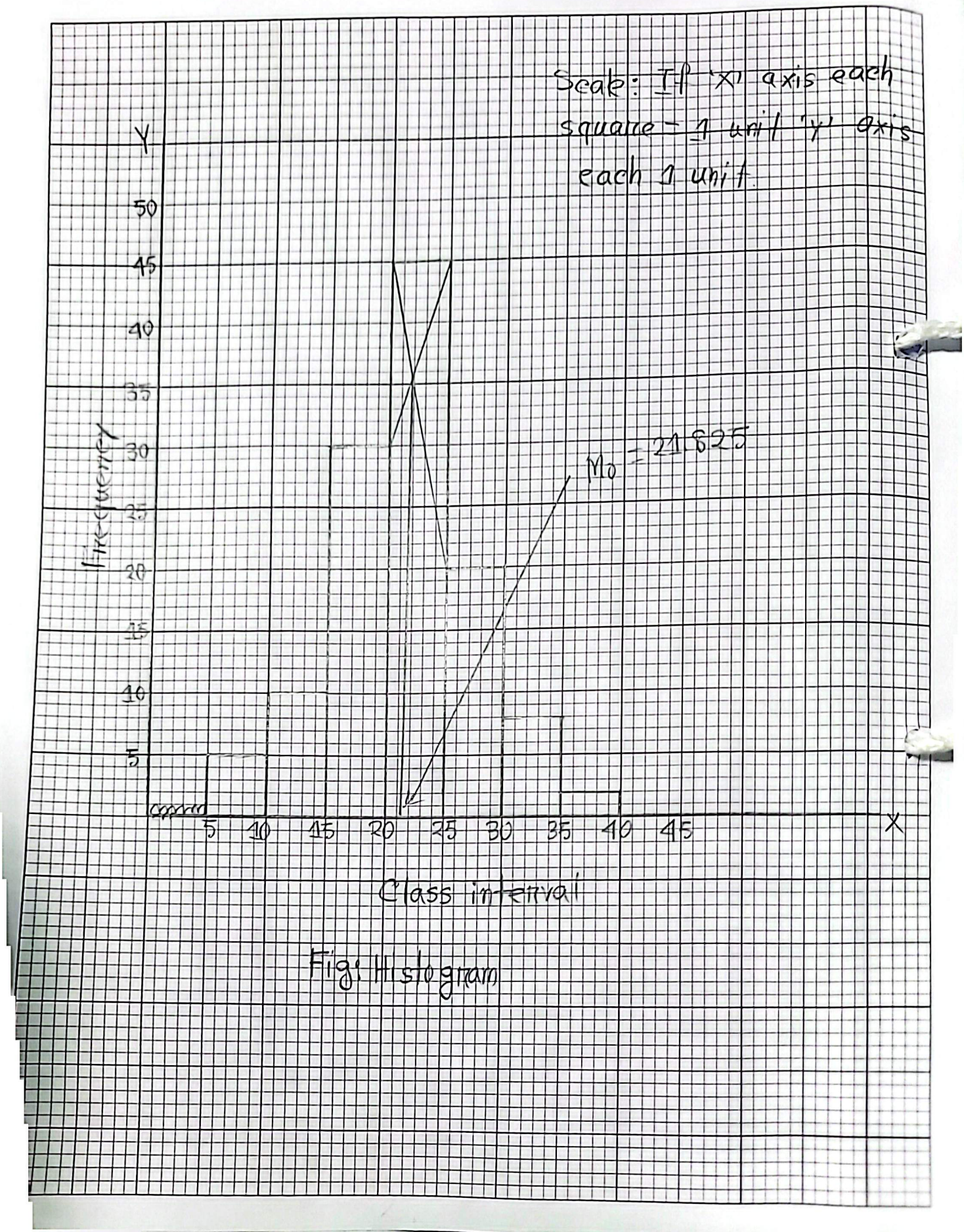


Fig: Histogram

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Comment: In the graph paper 'x' axis shows the class interval and each small square = 1 unit. The y axis shows the frequency and each small square = 1 unit

In the histogram show the Mode (M_o) is 21.875

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